

9(a)

any **four** from:

- 1 (blue) light decreases germination / AW ;
- 2 as intensity of (blue) light increases percentage germination decreases ;
- 3 data quote ;

light intensity / arbitrary units (au)	percentage germination
0 (dark)	98.0
36	76.9
48	45.0
57	14.7

- 4 (blue) light increases (the concentration of) ABA ;
- 5 ABA inhibits gibberellin (synthesis / activity) ;
- 6 any further valid suggestion of ABA action ;
e.g. DELLA not broken down / promotes dormancy / stops gene expression

4

Question	Answer	Marks
9(b)	any seven from: 1 auxin binds to receptor on the cell surface membrane ; 2 stimulates proton pumps ; 3 protons move from, cytoplasm / cell, to cell wall ; 4 pH of cell wall decreases ; 5 expansins activated ; 6 loosens linkage between cellulose microfibrils ; 7 by breaking hydrogen bonds ; Ignore weakening bonds 8 K⁺ channels open ; 9 K⁺ diffuse into, cytoplasm / cell ; 10 water potential (of cytoplasm / cell) decreases ; 11 water enters by osmosis ; 12 increase in turgor pressure / volume of cell increases ;	7